

2	(a) 1.	constant velocity / speed	B1	[1]
	2.	<i>either</i> constant / uniform decrease (in velocity/speed) <i>or</i> constant rate of decrease (in velocity/speed)	B1	[1]
	(b) (i)	distance is area under graph for both stages stage 1: distance $(18 \times 0.65) = 11.7$ (m) stage 2: distance $= (9 \times [3.5 - 0.65]) = 25.7$ (m) total distance $= 37.4$ m (-1 for misreading graph) {for stage 2, allow calculation of acceleration (6.32 m s^{-2}) and then $s = (18 \times 2.85) + \frac{1}{2} \times 6.32 (2.85)^2 = 25.7$ m}	C1	
	(ii)	<i>either</i> $F = ma$ $a = (18 - 0)/(3.5 - 0.65)$ <i>or</i> $E_K = \frac{1}{2}mv^2$ $E_K = \frac{1}{2} \times 1250 \times (18)^2$	C1 C1	
		$F = 1250 \times 6.3 = 7900$ N <i>or</i> $F = \frac{1}{2} \times 1250 \times (18)^2 / 25.7 = 7900$ N <i>or</i> initial momentum $= 1250 \times 18$ $F = \text{change in momentum} / \text{time taken}$ $F = (1250 \times 18) / 2.85 = 7900$	A1 (C1) (C1) (A1)	[3]
	(c) (i)	stage 1: <i>either</i> half / less distance as speed is half / less <i>or</i> half distance as the time is the same <i>or</i> sensible discussion of reaction time	B1	[1]
	(ii)	stage 2: <i>either</i> same acceleration and $s = v^2 / 2a$ <i>or</i> v^2 is $\frac{1}{4}$ $\frac{1}{4}$ of the distance	B1 B1	[2]

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3 (a) (i) power = work done per unit time / energy transferred per unit time / rate of work